

THE TESTS YOU ARE MOST LIKELY TO SIT

Test	Publisher	Items / time
Bennett Mechanical Comprehension Test II (BMCT-II)	Pearson TalentLens	55 items, 25 minutes
Wiesen Test of Mechanical Aptitude (WTMA)	Criteria Corp	60 items, 30 minutes
Ramsay Mechanical Aptitude Test (Form MAT-5)	Ramsay Corporation	36 items, 20 minutes
Employer or union batteries	Varies	Confirm the length in your invitation letter

WHAT THEY ALL HAVE IN COMMON

Question style	Mostly short mechanical-reasoning items, commonly using drawings or diagrams with a small set of answer choices; presentation varies by test
Knowledge needed	Everyday physical reasoning, not engineering maths — very little arithmetic
Ramsay MAT content	Household objects; work in production and maintenance; school science and physics; hand and power tools
Pace	Roughly 25 to 35 seconds per item on every one of these tests
Scoring	Varies by publisher; norm-referenced or standardised scores are common. Current online BMCT-II scoring also accounts for item difficulty rather than number correct alone
Calculator	Assume none unless your invitation says otherwise
Guessing	No penalty on the standard published forms — answer every item unless told otherwise

THE PRINCIPLES BEHIND MOST ITEMS

Lever	$F_1 \times d_1 = F_2 \times d_2$ Distances are measured from the fulcrum
Mechanical advantage	$MA = \text{load} \div \text{effort}$ Any MA above 1 trades distance for force
Fixed pulley	changes direction only, $MA = 1$ Effort equals the load
Movable pulley	$MA = \text{strands supporting the load}$ Two strands halve the effort, double the rope
Gear ratio	$N_1 \times T_1 = N_2 \times T_2$ $N = \text{speed}, T = \text{teeth}$
Gear direction	meshed gears turn opposite ways An idler changes the output direction without changing the overall driver-to-driven speed ratio
Belt drive	open belt same way, crossed belt reversed Small pulley spins faster
Pressure	$P = F \div A$ Smaller area under the same force = higher pressure
Hydraulic press	$F_1 \div A_1 = F_2 \div A_2$ Large piston lifts more but travels less
Inclined plane	$MA = \text{length} \div \text{height}$ Longer ramp, less force, more distance
Screw	finer pitch gives greater ideal MA, but more turns A longer, thinner wedge likewise gives greater ideal MA
Work and power	$W = F \times d; P = W \div t$ An ideal machine conserves work; a real one needs extra input because of losses
Density	objects float when less dense than the fluid Shape matters only through displacement
Speed of a wheel	$\text{rim speed} = \pi d \times \text{revolutions per minute}$ Bigger wheel covers more ground per turn
Spring stiffness	in series softer, in parallel stiffer Parallel springs share the load
Suspended load	its centre of gravity settles directly below the support point Basis of most beam and sling items

TIP These are aptitude tests, not knowledge tests. You cannot memorise the item bank — but you can make the dozen principles above automatic, which is what the timing actually rewards.



GEARS, PULLEYS AND BELTS

- Count the meshes, not the wheels: direction flips at every mesh in a gear train.
- A small gear driving a large one gives slower speed and more torque; the reverse speeds it up.
- Chain and open-belt drives keep both wheels turning the same way; a crossed belt reverses the driven wheel.
- In a block and tackle, count the rope sections actually pulling up on the moving block.
- Whatever you gain in force you lose in distance — that is the fastest sanity check on any answer.

BALANCE, LEVERS AND STRUCTURES

- A load hangs directly under its support point, and a body tips once its centre of gravity passes outside its base.
- Wider base and lower centre of gravity means more stable — this settles most "which will tip" items.
- To keep a beam level, hook it above the centre of gravity; move the hook toward the heavier end.
- Ideal second-class levers have force mechanical advantage greater than 1; ideal third-class levers less than 1. Classify from the positions of the fulcrum, effort and load, not from the object's name.
- Triangles are geometrically rigid; a rectangular frame is stabilised by diagonal bracing. A cable diagonal carries tension only, while a rigid brace may carry tension or compression.

FLUIDS, HEAT AND MOTION

- Liquid pressure depends on depth only — container width and shape do not change it.
- A liquid finds the same level in all connected branches, whatever their size.
- For steady incompressible flow a smaller flow area raises local velocity, and static pressure generally drops through the restriction.
- Heat travels from hot to cold; metals conduct fastest, still air is a good insulator.
- Materials expand when heated, which is why rails, pipe runs and bridges need gaps.
- Heavier rotating parts carry more inertia — slower to start, slower to stop, smoother once running.
- Friction grows with the force pressing the surfaces together and falls with rollers, wheels or lubricant.

TRAPS THAT COST POINTS

- Answering from the picture's appearance instead of the labelled part the question names.
- Choosing "equal" from appearance: outcomes match only when every relevant force, dimension and constraint is symmetrical.
- Mixing up speed and force — a gear change cannot increase both.
- Reading a pulley diagram as a movable pulley when it is anchored and merely redirects the rope.
- Assuming a bigger container means more pressure at the bottom.
- Counting pulley wheels instead of the rope strands that actually support the load.

TEST-DAY PLAN

1. Answer in order and keep moving; these tests are scored on how many you finish correctly, not on how hard you thought.
2. Give yourself about 30 seconds an item. If you are still reasoning at 45 seconds, guess and move.
3. Cover the answer choices, work out what should happen, then look — it stops plausible distractors from steering you.
4. Sketch the rope, gear or lever in the margin when one is allowed; two seconds of drawing beats twenty of staring.
5. On the standard published BMCT-II, WTMA and Ramsay MAT forms, answer every item; follow the instructions if your employer uses a customised battery.
6. Practise against a clock. Most candidates know the physics and lose points purely to pace.
7. Read the question's last line twice: "which takes more force" and "which moves faster" usually have opposite answers.
8. Expect a norm-referenced score, so finishing matters — unanswered items are simply lost marks.

TIP If two answers still look possible, pick the one that obeys conservation of energy: no arrangement of gears, pulleys or ramps ever gives out more work than is put into it.

